DEPARTMENT OF PHYSICS
INDIAN INSTITUTE OF TECHNOLOGY, MADRAS

PH1020 Physics II

Problem Set 7

emf, Electromagnetic induction, and Maxwell's equations Date: 15 Apr 2025

1. A square loop of wire with side length a lies at a distance s from a very long straight wire carrying current I. Assume the loop and the wire on the same plane.
   (a) Find the magnetic flux through the loop.
   (b) When the loop moves directly away from the wire, at a speed v, calculate the emf generated. Also indicate the direction of the current flow in the loop.
   (c) What will happen when the loop moves with speed v in the direction of the current flow?
   (d) Evaluate the mutual inductance between the wire and the loop.
   (e) If the current through the wire has a time-dependence given by $I = I_0 e^{-\lambda t}$ where I_0 and λ are positive constants, find the direction and the value of the emf induced in the square loop.
2. The magnetic field in an infinitely long cylindrical region is given by $\mathbf{B} = B_0 \hat{\boldsymbol{z}} \cos(\omega t + \alpha)$ for $\varrho \leq a$, and $\mathbf{B} = 0$ for $\varrho > a$. Here ϱ is the usual cylindrical polar coordinate, and B_0, ω, α and a are positive constants. Find the induced electric field at all points in space.
3. A toroidal coil of rectangular cross-section with inner radius a and outer radius b has height h and n turns. If a current I flows through its windings, find the magnetic flux Φ_m and hence the self-inductance of the toroid.
4. A small circular loop of wire (of radius a) lies at a distance z above the centre of a larger circular loop (of radius $b \gg a$). The planes of the loop are parallel to each other and perpendicular to the common axis of symmetry (see figure 1).

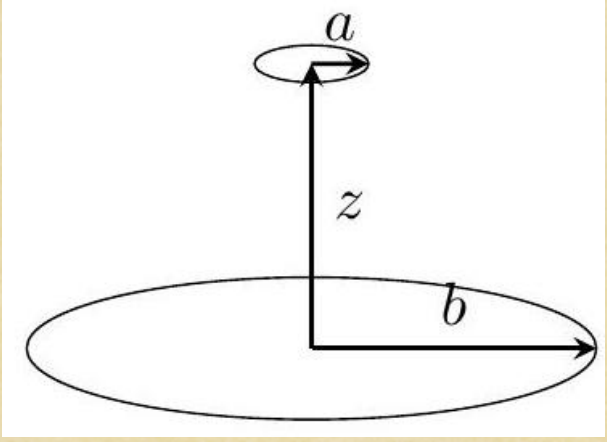


Figure 1: Figure for problem 4

(a) Suppose a current I flows in the larger loop. Determine the magnetic flux through the smaller loop. (Assume that the field across the smaller loop is uniform.)

(b) Suppose a current I flows in the small loop. Determine the magnetic flux through the big loop. (Assume the small loop as a point magnetic dipole.)

(c) Find the mutual inductance.

5. (a) Determine the self-inductance per unit length of a long coaxial cable of inner radius a and outer radius b. The region between the inner and the outer conductor is filled with a linear magnetic material with relative permeability μ_r.

(b) Calculate the energy stored (per unit length) in this inductor if a current I flows along the inner conductor and returns along the outer conductor.

(c) Calculate the power transported down the cables when both the conductors are held at a potential difference V.

6. The electric and magnetic fields are given by

$$\mathbf{E}(r,t) = -\frac{1}{4\pi\epsilon_0}\frac{1}{r^2}\theta(vt-r)\hat{\mathbf{r}}, \quad \mathbf{B}(r,t) = 0$$

Where

$$\theta(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Show that these fields satisfy the four Maxwell's equation. Find the charge density ρ and the current density $\mathbf{J}$. What kind of physical situation gives rise these fields?

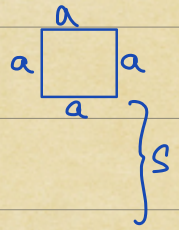
7. A transformer has two coils wrapped around an iron core in such a way that the same flux passes through every turn of both coils. The primary coil has n_1 turnsand the secondary coil has n_2 turns.

(a) If the current I_1 through the primary coil is changing, what is the ratio of the emfs in the primary and secondary coils.

(b) Show that, $M^2 = L_1 L_2$ where M is the mutual inductance and L_1, L_2 are individual self inductances.

8. Consider a long solenoid whose axis of symmetry is along z axis and a constant current I runs through it. A metal coin is allowed to fall under gravity along the axis of the solenoid. Find the induced emf and hence, the force on the coin due to the magnetic interaction when the coin is at a distance b from the top of the solenoid. (Neglect the mutual inductance for simplicity.)
(**Useful parameters:** radius of the coin is a, thickness of the coin is τ, conductivity of the coin is σ, cross-sectional area of the solenoid is A. You can neglect the mutual inductance)

1. A square loop of wire with side length a lies at a distance s from a very long straight wire carrying current I. Assume the loop and the wire on the same plane.
   (a) Find the magnetic flux through the loop.
   (b) When the loop moves directly away from the wire, at a speed v, calculate the emf generated. Also indicate the direction of the current flow in the loop.
   (c) What will happen when the loop moves with speed v in the direction of the current flow?
   (d) Evaluate the mutual inductance between the wire and the loop.
   (e) If the current through the wire has a time-dependence given by $I = I_0 e^{-\lambda t}$ where I_0 and λ are positive constants, find the direction and the value of the emf induced in the square loop.



(a) $\bar{B} = \frac{\mu_0 I}{2\pi r}$

$$\int \bar{B}\cdot d\bar{a} = \int_s^{s+a} \frac{\mu_0 I}{2\pi r}\, a\, dr$$

$$\phi = \frac{\mu_0 I a}{2\pi} \ln\left(\frac{s+a}{s}\right)$$

$$v = \frac{-ds}{dt}$$

(b) $$\mathcal{E} = \frac{-d\phi}{dt} = -\frac{\mu_0 I a}{2\pi} \frac{s}{s+a} \frac{a(-ds)}{s^2\, dt} = \frac{\mu_0 I a^2 v}{2\pi s(s+a)}$$

anticlockwise current

(c) No induced emf

(d) $M = \frac{\mu_0 a}{2\pi} \ln\left(\frac{s+a}{s}\right)$

(e) $\phi = \frac{\mu_0 I_0 e^{-\lambda t} a}{2\pi} \ln\left(\frac{s+a}{s}\right)$

$$\mathcal{E} = \frac{-d\phi}{dt} = \frac{\lambda \mu_0 I_0 e^{-\lambda t} a}{2\pi} \ln\left(\frac{s+a}{s}\right)$$ (anticlockwise current)

2. The magnetic field in an infinitely long cylindrical region is given by $\mathbf{B} = B_0\hat{\boldsymbol{z}}\cos(\omega t + \alpha)$ for $\varrho \leq a$, and $\mathbf{B} = 0$ for $\varrho > a$. Here ϱ is the usual cylindrical polar coordinate, and B_0, ω, α and a are positive constants. Find the induced electric field at all points in space.

$$\vec{\nabla}\times\vec{A} = \left[\frac{1}{\rho}\frac{\partial A_z}{\partial\varphi} - \frac{\partial A_\varphi}{\partial z}\right]\hat{e}_\rho + \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial\rho}\right]\hat{e}_\varphi + \frac{1}{\rho}\left[\frac{\partial}{\partial\rho}(\rho A_\varphi) - \frac{\partial A_\rho}{\partial\varphi}\right]\hat{e}_z$$

$$-\frac{dB}{dt} = B_0\omega\sin(\omega t+\alpha)\hat{z} \qquad s \leq a$$

$$= 0 \qquad s > a$$

$$\Rightarrow \bar{\nabla}\times\bar{E} = B_0\omega\sin(\omega t+\alpha)\hat{z} \qquad s \leq a$$

$$= 0 \qquad s > a$$

$$\Rightarrow \frac{1}{s}\frac{d}{ds}(sE\phi) = B_0\omega\sin(\omega t+\alpha) \qquad s \leq a$$

$$= 0 \qquad s > a$$

$$\Rightarrow E\phi = \frac{B_0 s\omega\sin(\omega t+\alpha)}{2} + c \qquad s \leq a$$

$$= d/s \qquad s > a$$

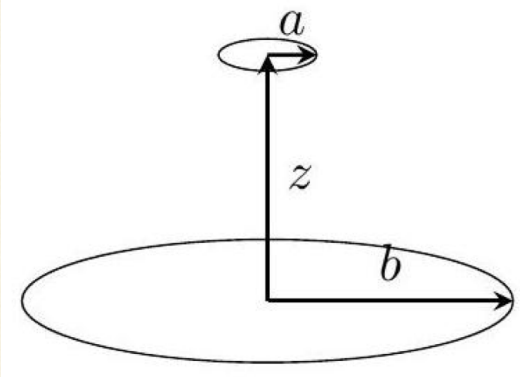
3. A toroidal coil of rectangular cross-section with inner radius a and outer radius b has height h and n turns. If a current I flows through its windings, find the magnetic flux Φ_m and hence the self-inductance of the toroid.

$$B = \frac{\mu_0 n i}{2\pi s}$$

$$\phi = \int\bar{B}\cdot d\bar{a} = \frac{\mu_0 n^2 i h}{2\pi}\ln(b/a) = Li$$

$$\Rightarrow L = \frac{\mu_0 n^2 h}{2\pi}\ln(b/a)$$

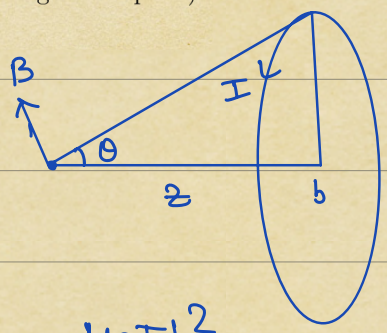
4. A small circular loop of wire (of radius a) lies at a distance z above the centre of a larger circular loop (of radius $b \gg a$). The planes of the loop are parallel to each other and perpendicular to the common axis of symmetry (see figure 1).



(a) Suppose a current I flows in the larger loop. Determine the magnetic flux through the smaller loop. (Assume that the field across the smaller loop is uniform.)

(b) Suppose a current I flows in the small loop. Determine the magnetic flux through the big loop. (Assume the small loop as a point magnetic dipole.)

(c) Find the mutual inductance.



(a) $dB = \frac{\mu_0}{4\pi} \frac{I\,dl}{(z^2+b^2)}$

$\int dB \sin\theta = \frac{\mu_0}{2\pi} \times \frac{I \times 2\pi b^2}{(z^2+b^2)^{3/2}} = \frac{\mu_0 I b^2}{2(z^2+b^2)^{3/2}}$

$\phi = \int B.da = \frac{\mu_0 I \pi a^2 b^2}{2(z^2+b^2)^{3/2}}$

(b) $\phi = \int \bar{A}.d\bar{l}$

$= \frac{\mu_0 I \pi a^2 b}{4\pi (z^2+b^2)^{3/2}} . 2\pi b$

$= \frac{\mu_0 I \pi a^2 b^2}{2(z^2+b^2)^{3/2}}$

$A = \frac{\mu_0}{4\pi} \frac{\bar{m} \times \hat{r}}{r^2}$

$\bar{m} = I \pi a^2 \hat{z}$

(c) $M = \dfrac{\mu_0 \pi a^2 b^2}{2(z^2+b^2)^{3/2}}$

5. (a) Determine the self-inductance per unit length of a long coaxial cable of inner radius a and outer radius b. The region between the inner and the outer conductor is filled with a linear magnetic material with relative permeability μ_r.
   (b) Calculate the energy stored (per unit length) in this inductor if a current I flows along the inner conductor and returns along the outer conductor.
   (c) Calculate the power transported down the cables when both the conductors are held at a potential difference V.

(a) $B \cdot 2\pi r = \mu_0 \mu_r I \Rightarrow B = \dfrac{\mu_0 \mu_r I}{2\pi r}$

$\phi = \int B \cdot dr \cdot dL$

$\Rightarrow \phi/l = \dfrac{\mu_0 \mu_r I}{2\pi} \ln(b/a)$

$\Rightarrow L/l = \dfrac{\mu_0 \mu_r \ln(b/a)}{2\pi}$

(b) $\frac{1}{2} \times L/l \times I^2 = \dfrac{\mu_0 \mu_r}{4\pi} \ln(b/a) I^2$

(c) $V = p \ln r + q$

$V(r=a) = V \qquad V = p \ln a + q$

$V(r=b) = 0 \qquad 0 = p \ln b + q$

$\Rightarrow \dfrac{V}{\ln(a/b)} = p$

$V = \dfrac{V \ln r}{\ln(a/b)} - \dfrac{\ln b}{\ln(a/b)} = \dfrac{V \ln(r/b)}{\ln(a/b)}$

$$\Rightarrow E = -\frac{dV}{d\gamma} = -\frac{V}{\ln(a/b)} \cdot \frac{b}{\gamma} \cdot \frac{1}{b} = \frac{V}{\ln(b/a)} \frac{1}{\gamma} \hat{\gamma}$$

$$\bar{B} = \frac{\mu_0 \mu_\gamma I}{2\pi\gamma} \hat{\phi}$$

$$\bar{S} = \frac{1}{\mu_0 \mu_\gamma} (\bar{E} \times \bar{B}) = \frac{I V}{2\pi\gamma^2 \ln(b/a)} \hat{z}$$

$$\vec{P} = \int \bar{S} \cdot d\bar{a} = \int_a^b \frac{IV}{2\pi\gamma^2 \ln(b/a)} \cdot 2\pi\gamma \, d\gamma = VI$$

6. The electric and magnetic fields are given by

$$\mathbf{E}(r,t) = -\frac{1}{4\pi\epsilon_0}\frac{1}{r^2}\theta(vt-r)\hat{\mathbf{r}}, \quad \mathbf{B}(r,t) = 0$$

Where

$$\theta(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Show that these fields satisfy the four Maxwell's equation. Find the charge density ρ and the current density $\mathbf{J}$. What kind of physical situation gives rise these fields?

$$\bar{\nabla} \cdot \bar{E} = \rho/\epsilon_0 \qquad \bar{\nabla} \cdot \bar{B} = 0 \qquad \bar{\nabla} \times \bar{E} = -\frac{\partial \bar{B}}{\partial t} \qquad \bar{\nabla} \times \bar{B} = \mu_0 (\bar{J}_f + \bar{J}_D)$$

$$\bar{J}_D = \epsilon_0 \frac{\partial \bar{E}}{\partial t}$$

$$\bar{\nabla} \cdot \bar{E} = \frac{1}{\gamma^2} \frac{d}{d\gamma} (\gamma^2 E) = \frac{1}{\gamma^2} \frac{d}{d\gamma} \left(\frac{-1}{4\pi\epsilon_0} \theta(ct-\gamma) \right)$$

$$= \frac{1}{4\pi\epsilon_0 \gamma^2} \delta(ct-\gamma) \cdot c$$

$$\Rightarrow \rho = \frac{\delta(ct-\gamma) \cdot c}{4\pi\gamma^2}$$

$\bar{\nabla} \cdot \bar{B} = 0$ ✓

$\bar{\nabla} \times \bar{B} = 0 = \mu_0 (\bar{J}) + \mu_0 \varepsilon_0 \frac{d\bar{E}}{dt}$

$\Rightarrow \mu_0 (J) + \mu_0 \varepsilon_0 \left(\frac{-1}{4\pi\varepsilon_0 r^2} \delta(vt - r) \cdot v \cdot v \right)$

$\Rightarrow J = \frac{v^2 \cdot \delta(vt - r)}{4\pi r^2} \hat{r}$ ✓

$\bar{\nabla} \times \bar{E} = \frac{-dB}{dt} = 0$ ✓

$J = \delta v \hat{r}$

7. A transformer has two coils wrapped around an iron core in such a way that the same flux passes through every turn of both coils. The primary coil has n_1 turnsand the secondary coil has n_2 turns.
   (a) If the current I_1 through the primary coil is changing, what is the ratio of the emfs in the primary and secondary coils.
   (b) Show that, $M^2 = L_1 L_2$ where M is the mutual inductance and L_1, L_2 are individual self inductances.

(a) $\mathcal{E}_1 = -n_1 \frac{d\phi}{dt}$ $\Rightarrow$ $\mathcal{E}_1 / \mathcal{E}_2 = n_1 / n_2$

$\mathcal{E}_2 = -n_2 \frac{d\phi}{dt}$ $\Rightarrow$

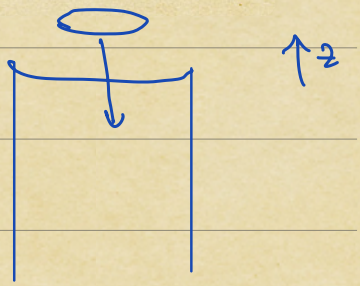
(b) $\phi_B(\text{coil})_1 = L_1 I_1 + M I_2 \Rightarrow \frac{L_1 I_1 + M I_2}{n_1} = \frac{L_2 I_2 + M I_1}{n_2}$

$\phi_B(\text{coil})_2 = L_2 I_2 + M I_1$ $\quad I_2 = 0 \Rightarrow M^2 = L_1 L_2$

8. Consider a long solenoid whose axis of symmetry is along z axis and a constant current I runs through it. A metal coin is allowed to fall under gravity along the axis of the solenoid. Find the induced emf and hence, the force on the coin due to the magnetic interaction when the coin is at a distance b from the top of the solenoid. (Neglect the mutual inductance for simplicity.)
(**Useful parameters:** radius of the coin is a, thickness of the coin is τ, conductivity of the coin is σ, cross-sectional area of the solenoid is A. You can neglect the mutual inductance)

$\Phi = B(z)\pi a^2$

$\Rightarrow \mathcal{E} = \pi a^2 \frac{-dB(z)}{dt} = \pi a^2 \frac{dB}{dz} v$



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PH1020 Physics II
Problem Set 7 - Solutions

emf, Electromagnetic induction, and Maxwell's equations Date: 15 Apr 2025

1. Problem-1

Question: A square loop of wire with side length a lies at a distance s from a very long straight wire carrying current I. Assume the loop and the wire on the same plane.

(a) Find the magnetic flux through the loop.

(b) When the loop moves directly away from the wire, at a speed v, calculate the emf generated. Also indicate the direction of the current flow in the loop.

(c) What will happen when the loop moves with speed v in the direction of the current flow?

(d) Evaluate the mutual inductance between the wire and the loop.

(e) If the current through the wire has a time-dependence given by $I = I_0 e^{-\lambda t}$ where I_0 and λ are positive constants, find the direction and the value of the emf induced in the square loop.

Solution:

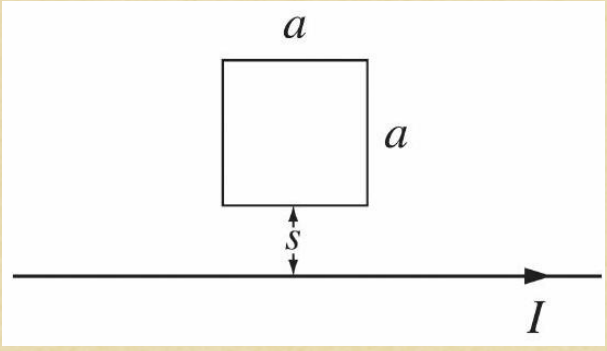


Figure 1: A square loop of side length a is kept at a distance s from an infinite wire carrying current I.

(a) Magnetic field at a distance s above the wire is

$$|\mathbf{B}| = \frac{\mu_0 I}{2\pi s}$$

in a direction out of the page. In order to find out the total flux through the square loop, let's first consider a thin rectangular strip at a distance y from the wire. The magnetic flux through the thin strip would be

$$d\Phi_B = \frac{\mu_0 I}{2\pi y} a dy$$

Therefore the net magnetic flux through the square loop is

$$\begin{aligned}\Phi_B &= \int d\Phi_B \\ &= \int_s^{s+a} \frac{\mu_0 I}{2\pi y} a dy \\ &= \frac{\mu_0 I a}{2\pi} \ln\left(\frac{s+a}{s}\right)\end{aligned}$$

(b) The induced emf

$$\begin{aligned}\mathcal{E} &= -\frac{d\Phi_B}{dt} \\ &= -\frac{d}{dt}\left(\frac{\mu_0 I a}{2\pi} \ln\left(\frac{s+a}{s}\right)\right) \\ &= -\frac{\mu_0 I a}{2\pi} \frac{s}{s+a}\left(\frac{-a}{s^2}\right)\frac{ds}{dt} \\ &= \frac{\mu_0 I a^2 v}{2\pi s(s+a)}\end{aligned}$$

The field value decreases as we move away from the wire so as the out of the page flux. The induced current would oppose the process and try to increase the out of the page flux. Therefore, the induced current will flow counterclockwise.

(c) When the loop moves parallel to the current flow in the wire the net magnetic flux does not change with time. Hence, the induced emf would be zero.

(d) From part (a) we have,

$$\Phi_B = \left(\frac{\mu_0 a}{2\pi} \ln\left(\frac{s+a}{s}\right)\right) I$$

Following equation (7.22) of [1], i.e. $\Phi_2 = M_{21} I_1$, we conclude the mutual inductance in this case,

$$M = \frac{\mu_0 a}{2\pi} \ln\left(\frac{s+a}{s}\right)$$

(e) The answer found above remains valid in the quasi-static case. [1] (Then Ampére's theorem can still be applied, leading to the relation $\oint \mathbf{B}\cdot\mathbf{dl} = \mu_0 I_{\text{enclosed}}$) Assuming this to be the case, emf induced in the loop,

$$\begin{aligned}&\mathcal{E} = -\frac{d\Phi_m}{dt} = -M\frac{dI}{dt}, \quad \text{or} \\ &\mathcal{E} = M\lambda I_0 e^{-\lambda t} \quad \text{(with } M \text{ as given above)}\end{aligned}$$

Now, B decreases as t increases. Hence, by Lenz's law, the direction of the induced current in the loop will be such that the field it creates will oppose this decrease in flux. Hence, current is in the direction shown by the double arrows in the figure above.

2. Problem-2

Question: The magnetic field in an infinitely long cylindrical region is given by $\mathbf{B} = B_0\hat{\boldsymbol{z}}\cos(\omega t + \alpha)$ for $\varrho \leq a$, and $\mathbf{B} = 0$ for $\varrho > a$. Here ϱ is the usual cylindrical polar coordinate, and B_0, ω, α and a are positive constants. Find the induced electric field at all points in space.

Solution:

Since $\mathbf{B}$ is spatially uniform inside the cylindrical region $\varrho \leq a$, but oscillates in time, and $\mathbf{B} = 0$ for $\varrho > a$, we can visualize this field as being produced by an alternating current (AC) flowing in an ideal infinite solenoid of radius a. By cylindrical symmetry, therefore, we may expect $\mathbf{E}$ to be of the form $\mathbf{E}(\varrho, \varphi, z) = E_\varphi(\varrho)\hat{\varphi}$. That is, it has only an azimuthal component, whose magnitude depends only on the radial distance ρ from the z-axis. Integrate over a (planar) circular path of radius ρ about the z-axis to obtain

$$\begin{aligned}\oint_C \mathbf{E}\cdot\mathbf{dl} &= \int_{\varphi=0}^{2\pi} \mathbf{E}_\varphi(\varrho)\hat{\varphi}\cdot\varrho d\varphi\hat{\varphi} \\ &= 2\pi\varrho E_\varphi(\varrho).\end{aligned}$$

But $\oint_C \mathbf{E}\cdot\mathbf{dl} = \int_S \boldsymbol{\nabla}\times\mathbf{E}\cdot\mathbf{da} = -\int_S \frac{\partial\mathbf{B}}{\partial t}\cdot\mathbf{da}$.
For $\rho \leq a$, (note that da is along $\hat{\mathbf{z}}$)

$$-\int_S \frac{\partial\mathbf{B}}{\partial t}\cdot\mathbf{da} = \omega B_0\sin(\omega t+\alpha)\pi\varrho^2$$

For $\rho > a$,

$$-\int_S \frac{\partial\mathbf{B}}{\partial t}\cdot\mathbf{da} = \omega B_0\sin(\omega t+\alpha)\pi a^2$$

Therefore

$$\mathbf{E}(\varrho,\varphi,z) = \begin{cases} \frac{\omega B_0}{2}\varrho\sin(\omega t+\alpha)\hat{\varphi}, & \text{for } \varrho \leq a \\ \frac{\omega B_0}{2}\frac{a^2}{\varrho}\sin(\omega t+\alpha)\hat{\varphi}, & \text{for } \varrho > a \end{cases}$$

3. Problem-3

Question: A toroidal coil of rectangular cross-section with inner radius a and outer radius b has height h and n turns. If a current I flows through its windings, find the magnetic flux Φ_m and hence the self-inductance of the toroid.

Solution:

The field $\mathbf{H}$ at a distance $\varrho(a \leq \varrho \leq b)$ from the z-axis is $\mathbf{H} = \frac{nI}{2\pi\varrho}\hat{\varphi}$.

$$\therefore \mathbf{B} = \frac{\mu_0 nI}{2\pi\varrho}\hat{\varphi}.$$

[01] This happens when the term $\mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t}$ in the Maxwell equation, that we are yet to discuss in the class lectures(!), $\boldsymbol{\nabla}\times\mathbf{B} = \mu_0\mathbf{J} + \mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t}$, can be neglected compared to the term $\mu_0\mathbf{J}$.

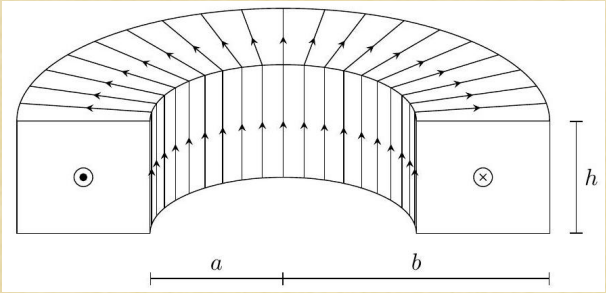


Figure 2: Toroidal Coil

The flux through

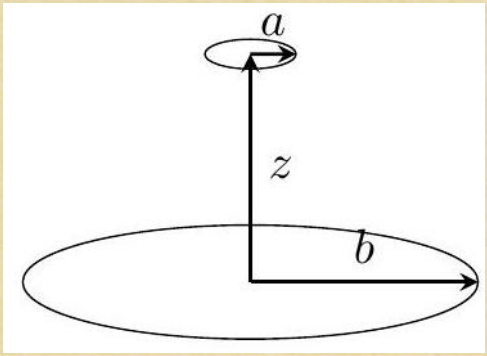
$$\Phi_m = n \int \mathbf{B} \cdot \mathbf{da}$$
$$= \frac{\mu_0 n^2 h I}{2\pi} \ln\left(\frac{b}{a}\right) \equiv LI.$$

Therefore, the self-inductance is given by

$$L = \frac{\mu_0 n^2 h}{2\pi} \ln\left(\frac{b}{a}\right)$$

4. Problem-4

Question: A small circular loop of wire (of radius a) lies at a distance z above the centre



of a larger circular loop (of radius $b \gg a$). The planes of the loop are parallel to each other and perpendicular to the common axis of symmetry (see figure **??**).

(a) Suppose a current I flows in the larger loop. Determine the magnetic flux through the smaller loop. (Assume that the field across the smaller loop is uniform.)

(b) Suppose a current I flows in the small loop. Determine the magnetic flux through the big loop. (Assume the small loop as a point magnetic dipole.)

(c) Find the mutual inductance.

Solution:

(a) The magnetic field due to the larger coil (call it coil 1) at the location of the small coil (call it coil 2) is

$$\mathbf{B}_1 = \frac{\mu_0 I_1 b^2}{2\left(b^2+z^2\right)^{3/2}}\hat{\mathbf{z}}$$

The flux $\Phi_{2,1}$ through loop 2 due to the current I_1 through loop 1 is

$$\Phi_{2,1} = \left(\pi a^2\right)\frac{\mu_0 I_1 b^2}{2\left(b^2+z^2\right)^{3/2}} = M_{21} I_1$$

where $M_{21} = \frac{\mu_0 \pi a^2 b^2}{2(b^2+z^2)^{3/2}}$.

(b) The magnetic dipole moment of the small loop due to a current I_2 flowing through it is

$$\mathbf{m} = \pi a^2 I_2 \hat{e}_z$$

The magnetic field due to the dipole located at $(0,0,z)$ at a point $(x,y,0)$ located in the xy-plane is

$$\mathbf{B}_2 = \frac{\mu_0}{4\pi\left(\varrho^2+z^2\right)^{3/2}}((3\mathbf{m}\cdot\hat{n})\hat{n}-\mathbf{m})$$

where $\hat{n}$ is the unit vector along $(\varrho\hat{\varrho} - z\hat{\mathbf{z}})$. The flux $\Phi_{1,2}$ through loop 1 due to the current I_2 through loop 2 is

$$\begin{aligned}
\Phi_{1,2} &= \int_{\text{loop 1}} (\mathbf{B}_2\cdot\hat{\mathbf{z}})\,\varrho d\varrho d\varphi \\
&= \int \frac{\mu_0 m}{4\pi\left(\varrho^2+z^2\right)^{3/2}}\left(\frac{3z^2}{(\varrho^2+z^2)}-1\right)2\pi\varrho d\varrho \\
&= \frac{\mu_0 m b^2}{2\left(b^2+z^2\right)^{3/2}}
\end{aligned}$$

after carrying out the ϱ integration. We thus obtain

$$\Phi_{1,2} = \frac{\mu_0 a^2 b^2}{2\left(b^2+z^2\right)^{3/2}} I_2 = M_{12} I_2$$

where $M_{12} = \frac{\mu_0 \pi a^2 b^2}{2(b^2+z^2)^{3/2}}$.

(c) From the definition of mutual inductance, we see that

$$M = M_{12} = M_{21} = \frac{\mu_0 \pi a^2 b^2}{2\left(b^2+z^2\right)^{3/2}}.$$

We saw the symmetry from Neumann's formula for mutual inductance. The symmetry $M_{ij} = M_{ji}$ also follows from the definition of energy stored in a system of inductors analogous to what happened to a system of conductors.

5. Problem-5

Question:

(a) Determine the self-inductance per unit length of a long coaxial cable of inner radius a and outer radius b. The region between the inner and the outer conductor is filled with a linear magnetic material with relative permeability μ_r.

(b) Calculate the energy stored (per unit length) in this inductor if a current I flows along the inner conductor and returns along the outer conductor.

(c) Calculate the power transported down the cables when both the conductors are held at a potential difference V.

Solution:

(a) Let the cable runs along $\hat{\mathbf{z}}$. The inner radius is a and outer radius is b. We will use Ampere's law for $\mathbf{H}$ and then find B. From equation (6.20) of [1],

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\mathrm{f}_{enc}}$$

$$\mathbf{H} = \frac{I}{2\pi s}\hat{\boldsymbol{\phi}} \quad (a < s < b)$$

Therefore,

$$\mathbf{B} = \mu_0\mu_r\mathbf{H} = \mu_0\mu_r\frac{I}{2\pi s}\hat{\boldsymbol{\phi}} \quad (a < s < b)$$

Magnetic flux per unit length,

$$\begin{aligned}\frac{d\Phi_B}{dz} &= \mu_0\mu_r\frac{I}{2\pi}\int_a^b \frac{1}{s}ds \\ &= \frac{\mu_0\mu_r}{2\pi}\ln\frac{b}{a}I\end{aligned}$$

Therefore, self inductance per unit length,

$$L = \frac{\mu_0\mu_r}{2\pi}\ln\frac{b}{a}$$

(b) Energy stored per unit length,

$$E = \frac{1}{2}LI^2 = \frac{\mu_0\mu_r}{4\pi}\ln\left(\frac{b}{a}\right)I^2$$

(c) The power (P) transported down the cables when both the conductors are held at a potential difference V can be written as,

$$P = VI$$

6. Problem-6

Question: The electric and magnetic fields are given by

$$\mathbf{E}(r,t) = -\frac{1}{4\pi\epsilon_0}\frac{1}{r^2}\theta(vt-r)\hat{\mathbf{r}}, \quad \mathbf{B}(r,t) = 0$$

Where

$$\theta(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Show that these fields satisfy the four Maxwell's equation. Find the charge density ρ and the current density $\mathbf{J}$. What kind of physical situation gives rise these fields?
Solution:

It is given,

$$\mathbf{E}(r,t) = -\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\theta(vt-r)\hat{\mathbf{r}}, \quad \mathbf{B}(r,t) = 0$$

Let's verify the Maxwell's equation one by one and determine the corresponding charge density, ρ and current density, $\mathbf{J}$.

- $\boldsymbol{\nabla}\cdot\mathbf{E} = \frac{\rho}{\epsilon_0}$

$$\begin{aligned}\boldsymbol{\nabla}\cdot\mathbf{E} &= -\frac{q}{4\pi\epsilon_0}\boldsymbol{\nabla}\cdot\left(\frac{r^2}{r^2}\theta(vt-r)\hat{\mathbf{r}}\right) \\ &= -\frac{q}{4\pi\epsilon_0}\left[\theta(vt-r)\boldsymbol{\nabla}\cdot\left(\frac{\hat{\mathbf{r}}}{r^2}\right) + \frac{\hat{\mathbf{r}}}{r^2}\cdot\boldsymbol{\nabla}(\theta(vt-r))\right] \\ &= -\frac{q\delta^3(\mathbf{r})}{\epsilon_0}\theta(vt-r) + \frac{q\delta(vt-r)}{4\pi\epsilon_0 r^2}\end{aligned}$$

Therefore, $\rho(\mathbf{r}) = -q\delta^3(\mathbf{r})\theta(vt-r) + \frac{q\delta(vt-r)}{4\pi r^2}$

- $\boldsymbol{\nabla}\cdot\mathbf{B} = 0$

This is trivial as $\mathbf{B}(r,t) = 0$

- $\boldsymbol{\nabla}\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}$

$$\boldsymbol{\nabla}\times\mathbf{E} = 0 = -\frac{\partial\mathbf{B}}{\partial t}$$

- $\boldsymbol{\nabla}\times\mathbf{B} = \mu_0\mathbf{J} + \mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t}$

It is obvious to see that, $\boldsymbol{\nabla}\times\mathbf{B} = 0$. We will use this to find the current distribution.

$$\begin{aligned}\frac{\partial\mathbf{E}}{\partial t} &= \frac{\partial}{\partial t}\left(\frac{-q}{4\pi\epsilon_0}\frac{1}{r^2}\theta(vt-r)\hat{\mathbf{r}}\right) \\ &= \frac{-q}{4\pi\epsilon_0}\frac{\hat{\mathbf{r}}}{r^2}\delta(vt-r)v\end{aligned}$$

[01] In spherical polar coordinate $\boldsymbol{\nabla}\times\mathbf{A} = \frac{1}{r\sin\theta}\left(\frac{\partial}{\partial\theta}\left(A_\phi\sin\theta\right) - \frac{\partial A_\theta}{\partial\phi}\right)\hat{\mathbf{r}} + \frac{1}{r}\left(\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{\partial}{\partial r}\left(rA_\phi\right)\right)\hat{\boldsymbol{\theta}} + \frac{1}{r}\left(\frac{\partial}{\partial r}\left(rA_\theta\right) - \frac{\partial A_r}{\partial\theta}\right)\hat{\phi}$

Therefore,

$$\mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = 0$$
$$\mathbf{J} = \frac{q}{4\pi r^2} v\delta(vt - r)\hat{\mathbf{r}}$$

7. Problem-7

Question: A transformer has two coils wrapped around an iron core in such a way that the same flux passes through every turn of both coils. The primary coil has n_1 turnsand the secondary coil has n_2 turns.

(a) If the current I_1 through the primary coil is changing, what is the ratio of the emfs in the primary and secondary coils.

(b) Show that, $M^2 = L_1 L_2$ where M is the mutual inductance and L_1, L_2 are individual self inductances.

Solution:

(a) Let Φ_B be the magnetic flux due to $\mathbf{B}$ through each turn of either coil. Therefore,

$$\Phi_1 = n_1 \Phi_B, \quad \text{and } \Phi_2 = n_2 \Phi_B$$

and

$$\frac{\mathcal{E}_1}{\mathcal{E}_2} = \frac{-n_1 \frac{d\Phi_B}{dt}}{-n_2 \frac{d\Phi_B}{dt}} = \frac{n_1}{n_2}$$

(b) Suppose current I_1 and I_2 passes through the primary and secondary coil respectively. Then,

$$\Phi_1 = I_1 L_1 + I_2 M = n_1 \Phi_B$$
$$\Phi_2 = I_2 L_2 + I_1 M = n_2 \Phi_B$$

Therefore,

$$\Phi_B = I_1 \frac{L_1}{n_1} + I_2 \frac{M}{n_1} = I_2 \frac{L_2}{n_2} + I_1 \frac{M}{n_2}$$

In case $I_1 = 0$, we have $\frac{M}{n_1} = \frac{L_2}{n_2}$ and in case $I_2 = 0$ then, $\frac{L_1}{n_1} = \frac{M}{n_2}$. Dividing these two results we get $M^2 = L_1 L_2$

8. Problem-8

Question: Consider a long solenoid whose axis of symmetry is along z axis and a constant current I runs through it. A metal coin is allowed to fall under gravity along the axis of the solenoid. Find the induced emf and hence, the force on the coin due to the magnetic interaction when the coin is at a distance b from the top of the solenoid. (Neglect the mutual inductance for simplicity.)
(**Useful parameters:** radius of the coin is a, thickness of the coin is τ, conductivity of the coin is σ, cross-sectional area of the solenoid is A. You can neglect the mutual inductance)

Solution:

Let's first analyze the problem qualitatively. Consider the current in the solenoid is counter-clockwise when one look from the top. As the coin approaches the solenoid upward magnetic flux through the coin will increase. Hence, it will induce a current in the coin clockwise which in result exert an upward force opposite to the gravity. We can see it as a drag force and at the end we will be able to find exactly the drag coefficient.

The idea to approach the problem is as follows: we consider an infinitesimal ring at a radial distance r. We will find the induced emf on the ring and then the force on the same. Then we will integrate from 0 to the radius of the coin.
The induced emf in the infinitesimal ring at a distance r, (following equation (7.11) of Griffith's book)

$$\mathcal{E}_r = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l} = 2\pi r v(t) B_\rho(r, b)$$

where $v(t) = \frac{dz}{dt}$ (the relative speed of the coin w.r.t. the solenoid) and $B_\rho(r, t)$ is the radial component of the magnetic field due to the solenoid. Notice, the vertical component of the magnetic field due to the solenoid will not contribute to the induced emf.

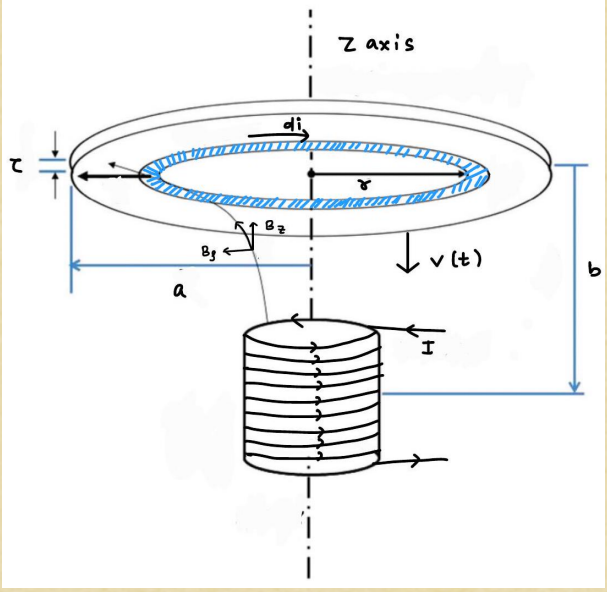


Figure 3: component of the magnetic field due to the solenoid will not contribute to the induced emf.

The radial component of the magnetic field can be written as :

$$B_\rho(r, b) = \frac{\mu_0 m}{4\pi} \frac{3rb}{(r^2 + b^2)^{5/2}}$$

where $m = NIA, N$ is the total turns in the solenoid. Now, the current in the ring due to this induced emf would be,

$dI = \mathcal{E}_r dG$ (dG is the conductance of the infinitesimal ring)

$$= (2\pi r v(t) B_\rho(r, b)) \left(\sigma\tau \frac{dr}{2\pi r} \right) \quad \left(dG = \sigma\tau \frac{dr}{2\pi r} \right)$$

$$= \mathrm{v(t)}\ \sigma\tau B_\rho(r, b) dr$$

The magnetic interaction force,

$$\begin{aligned}
F_z &= \int I(d\mathbf{l}\times\mathbf{B}) \\
&= 2\pi v(t)\sigma\tau \int_0^a B_\rho^2(r,b) r dr \\
&= 2\pi v(t)\sigma\tau \int_0^a \left(\frac{\mu_0 m}{4\pi}\frac{3rb}{(r^2+b^2)^{5/2}}\right)^2 r dr \\
&= 2\pi v(t)\sigma\tau \left(\frac{9b^2\mu_0^2 m^2}{16\pi^2}\right)\int_0^a \frac{r^3}{(r^2+b^2)^5} dr \qquad [\text{substitute } u = r/b] \\
&= \left[\frac{3}{4}\pi\sigma\tau\left(\frac{\mu_0 m}{4\pi}\right)^2 g\left(\frac{a}{b}\right)\right] v(t) \qquad \left[g(x) = 1 - \frac{1+4x^2}{(1+x^2)^4}\right]
\end{aligned}$$

Here you see $F \propto v$, which acts like a drag force. We can call $\frac{3}{4}\pi\sigma\tau\left(\frac{\mu_0 m}{4\pi}\right)^2 g\left(\frac{a}{b}\right)$ as the drag coefficient. So, the coin will fall under gravity with a drag force form the magnetic interaction. Given the numbers, one can work out the exact point where the coin stops. Students are encouraged to study the system considering the inductance. In that case, instead of a purely resistive circuit the coin will behave like an LR circuit. Further, one can study the dynamics for a sinusoidal current instead of a constant current through the solenoid.

[1]

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